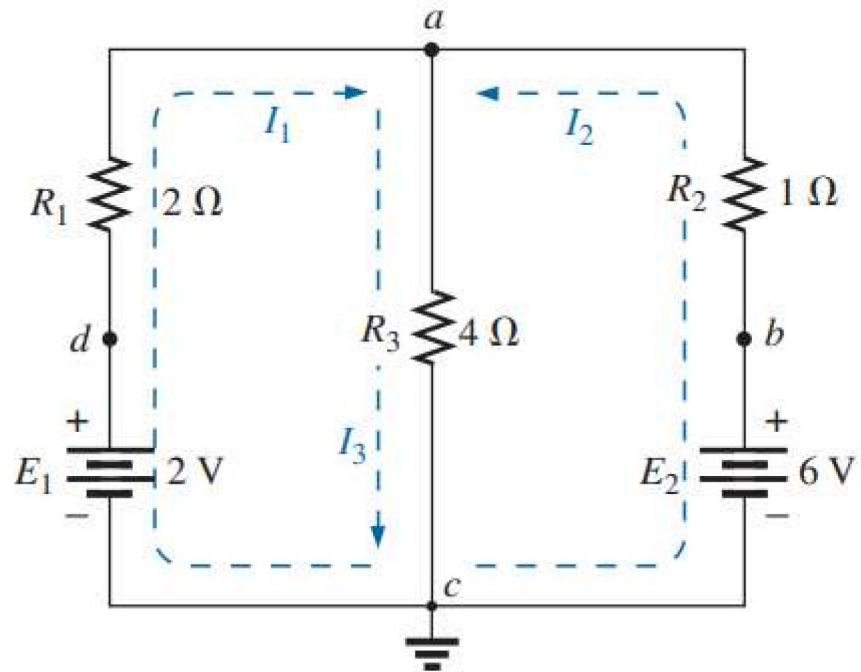


Lecture 10 & 11

Branch-Current Analysis Procedure

1. Assign a distinct current of arbitrary direction to each branch of the network.
2. Indicate the polarities for each resistor as determined by the assumed current direction.
3. Apply Kirchhoff's voltage law around each closed, independent loop of the network.
4. Apply Kirchhoff's current law at the minimum number of nodes that will include all the branch currents of the network.
5. Solve the resulting simultaneous linear equations for assumed branch currents.



- *18. a.** Using branch-current analysis, find the current through the $9.1\text{ k}\Omega$ resistor in Fig. 8.122. Note that all the resistors are standard values.
- b.** Using the results of part (a), determine the voltage V_a .

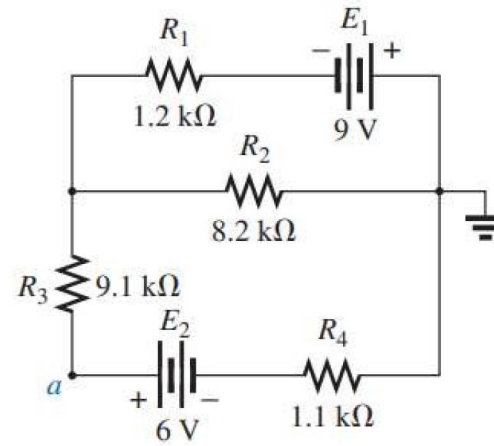
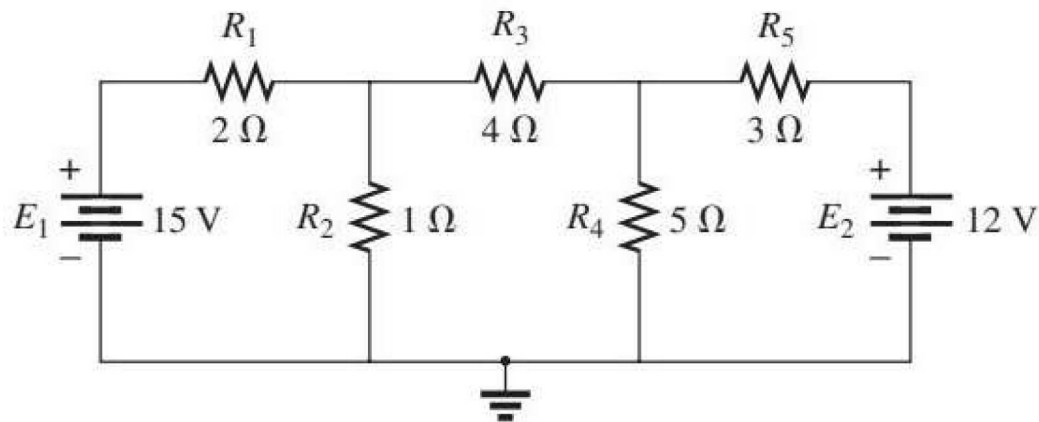


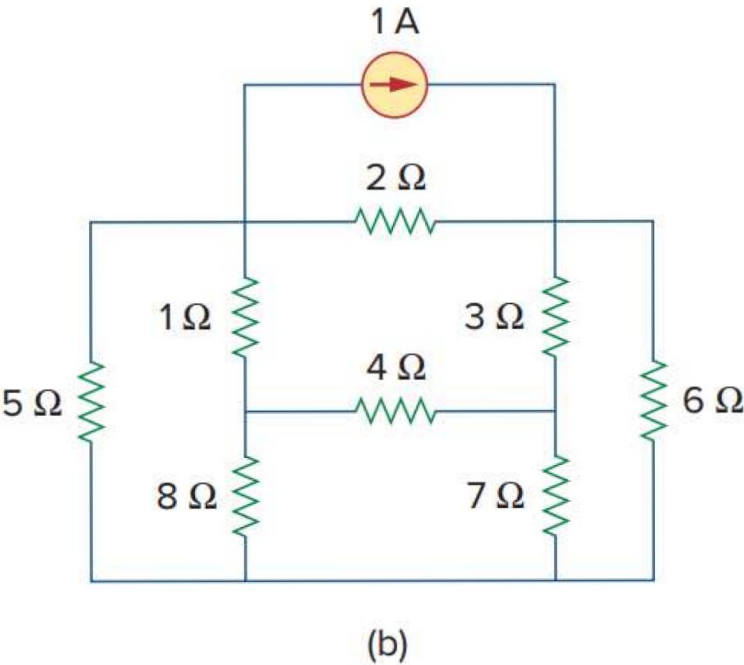
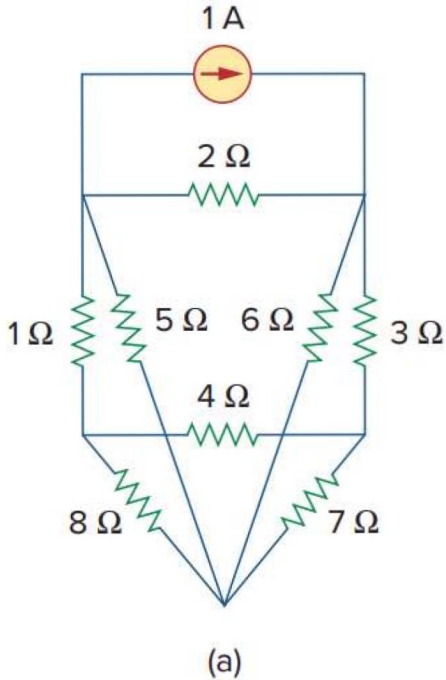
FIG. 8.122
Problems 18 and 23.

***19.** For the network in Fig. 8.123:

- Write the equations necessary to solve for the branch currents.
- By substitution of Kirchhoff's current law, reduce the set to three equations.
- Rewrite the equations in a format that can be solved using third-order determinants.
- Solve for the branch current through the resistor R_3 .



Networks that can be drawn on a plane without having one element (or conductor) crossing over another are called planar networks. On the other hand, circuits that must be drawn with one or more elements crossing others are said to be nonplanar. We consider only planar networks.



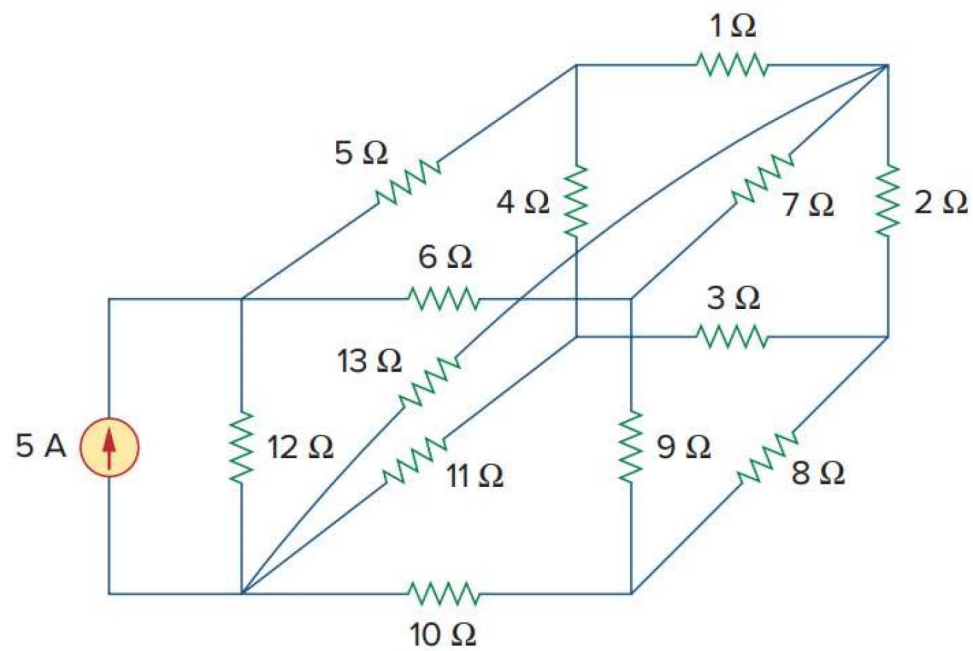
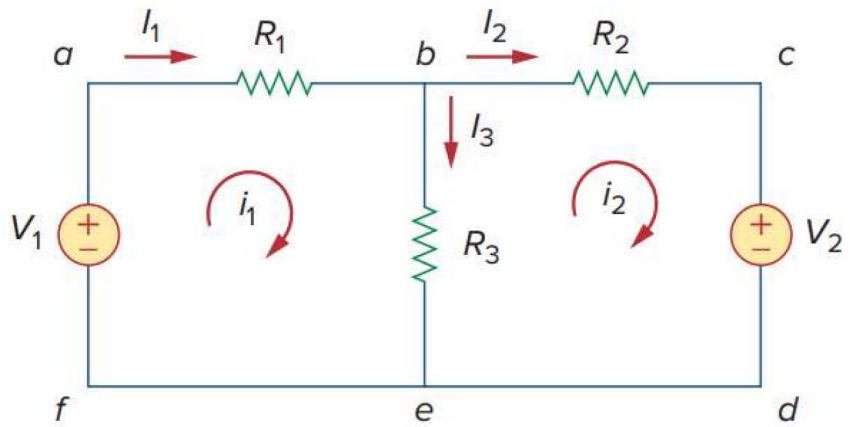


Figure 3.16
A nonplanar circuit.

A mesh is a loop that does not contain any other loops within it.



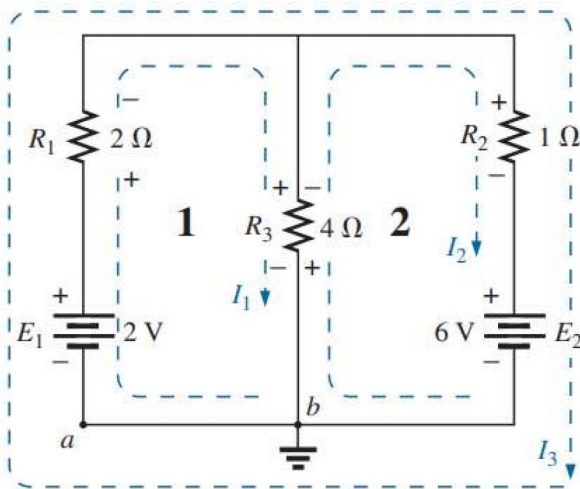
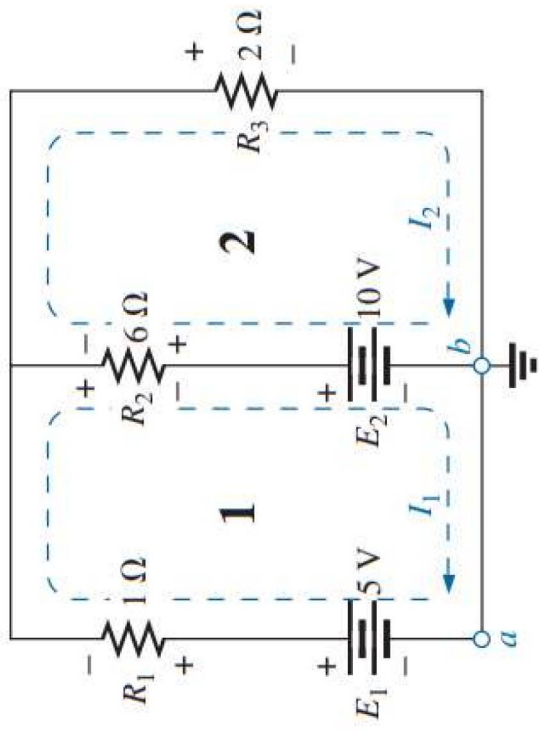
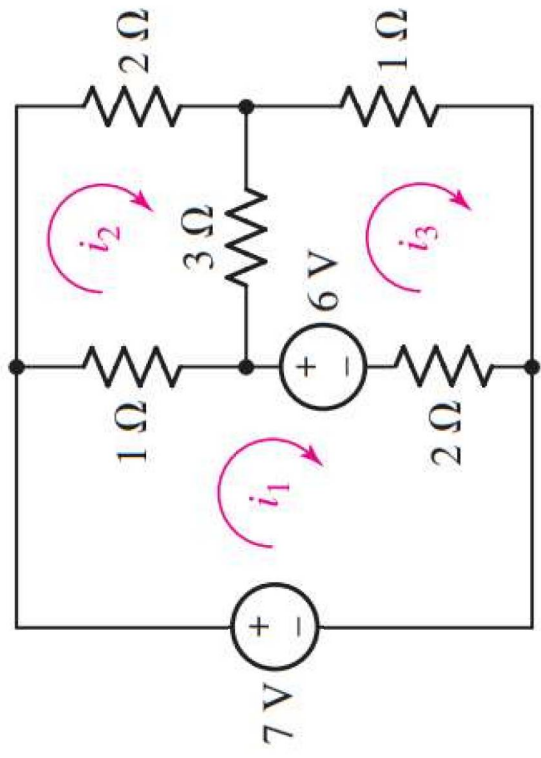
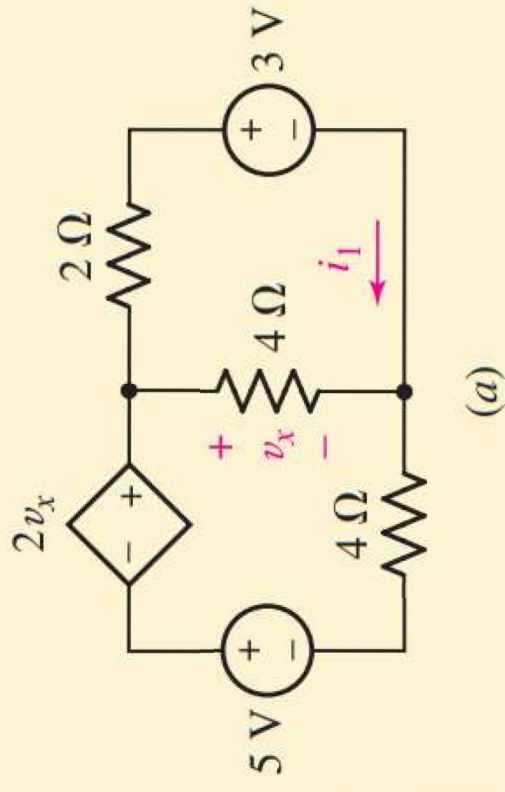


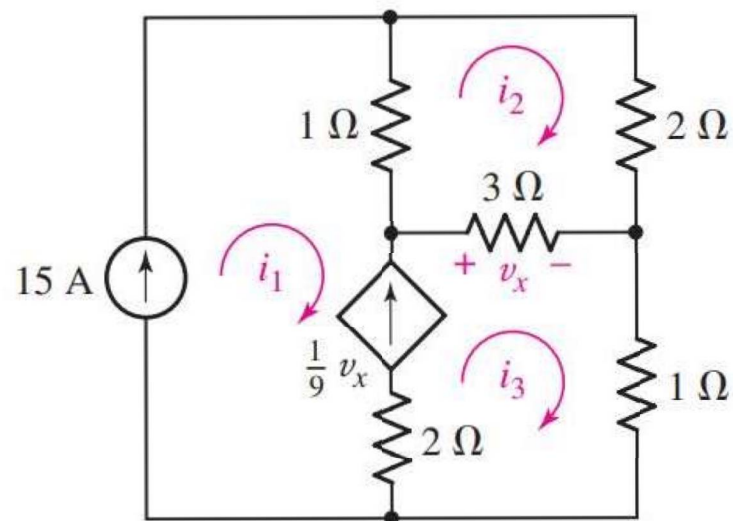
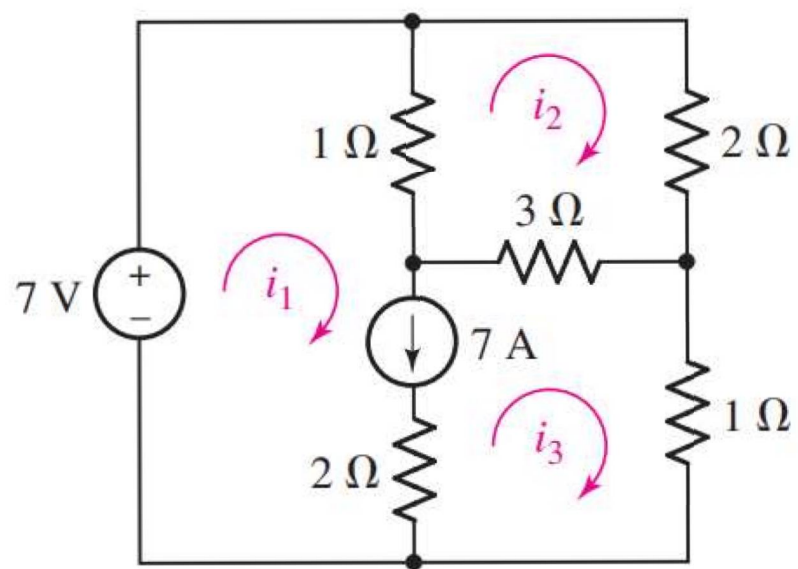
FIG. 8.35

*Defining the mesh currents for a
"two-window" network.*

1. indicate the polarities within each loop for each resistor as determined by the assumed direction of loop current for that loop. Note the requirement that the polarities be placed within each loop. This requires, as shown in Fig. 8.35, that the 4 Ω resistor have two sets of polarities across it.
2. If a resistor has two or more assumed currents through it, the total current through the resistor is the assumed current of the loop in which Kirchhoff's voltage law is being applied, plus the assumed currents of the other loops passing through in the same direction, minus the assumed currents through in the opposite direction.
3. The polarity of a voltage source is unaffected by the direction of the assigned loop currents.
4. If a current source lies on the periphery of a mesh, no KVL equation is needed and the mesh current is determined by inspection







■ **FIGURE 4.26** A three-mesh circuit with one dependent and one independent current source.

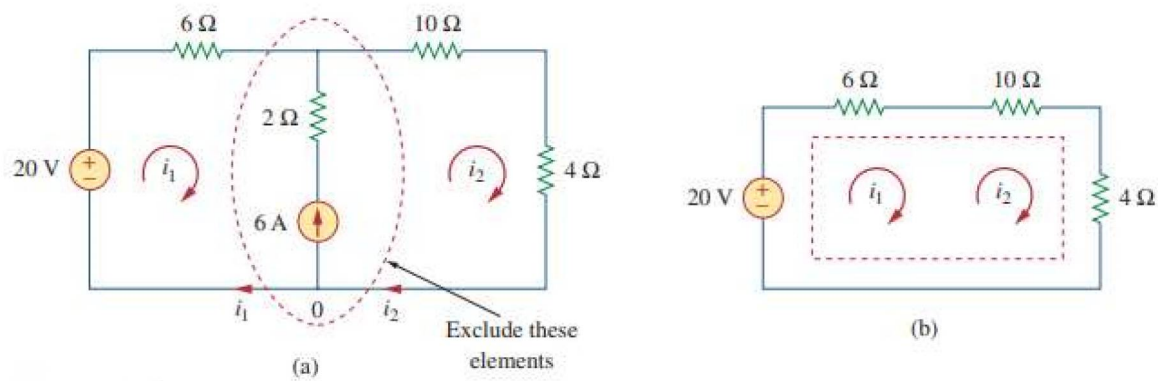
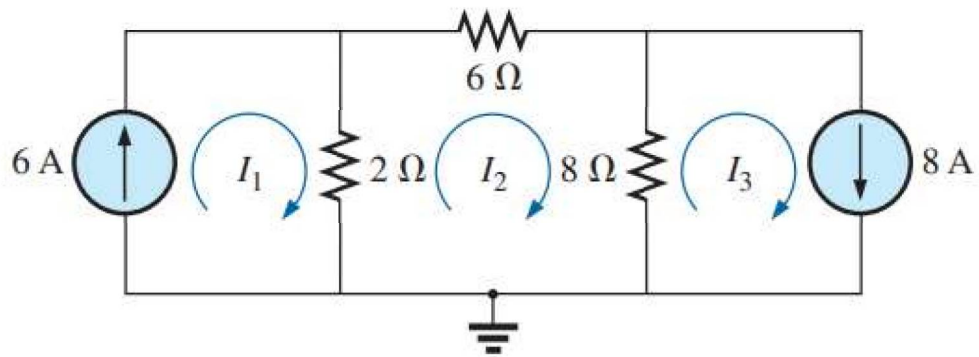


Figure 3.23

(a) Two meshes having a current source in common, (b) a supermesh, created by excluding the current source.



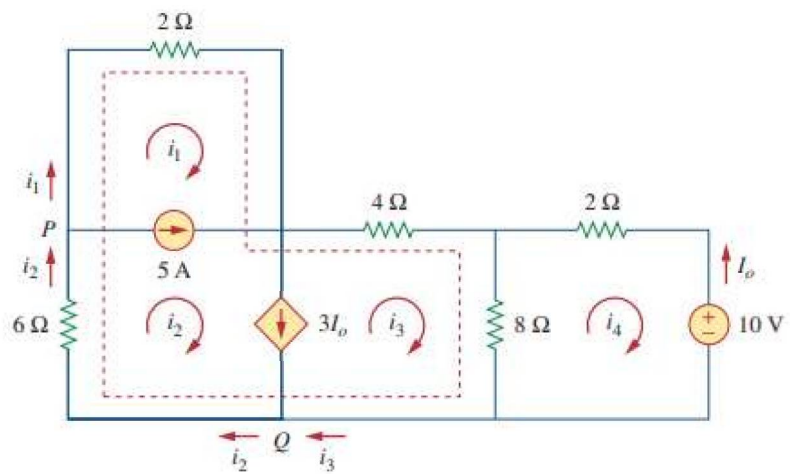


Figure 3.29
xample 3.9.

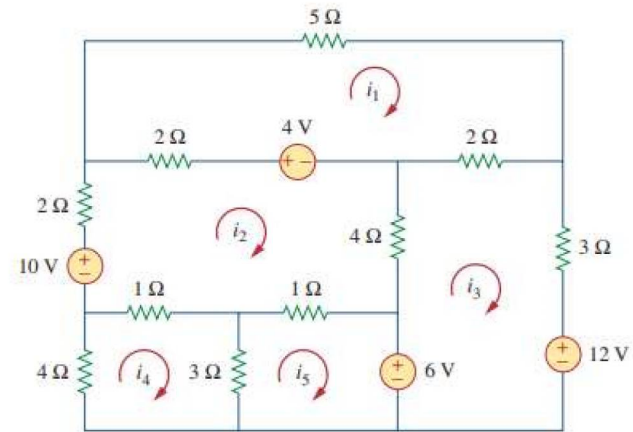


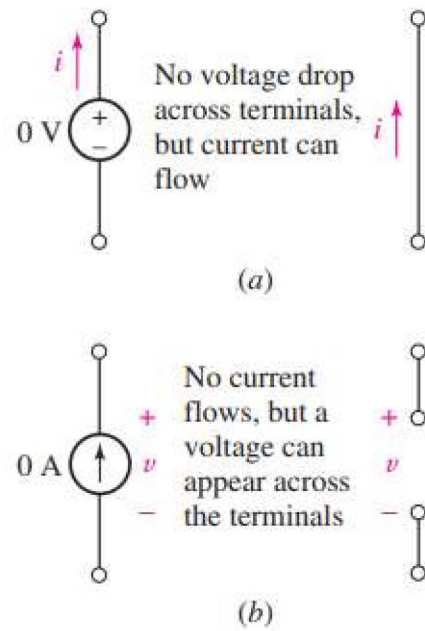
Figure 3.30
For Practice Prob. 3.9.

- A linear circuit is one whose output is linearly related (or directly proportional) to its input
- Linearity is the property of an element describing a linear relationship between cause and effect.
- The property is a combination of both the homogeneity (scaling) property and the additivity property.
- A linear circuit consists of only linear elements, linear dependent sources, and independent sources.

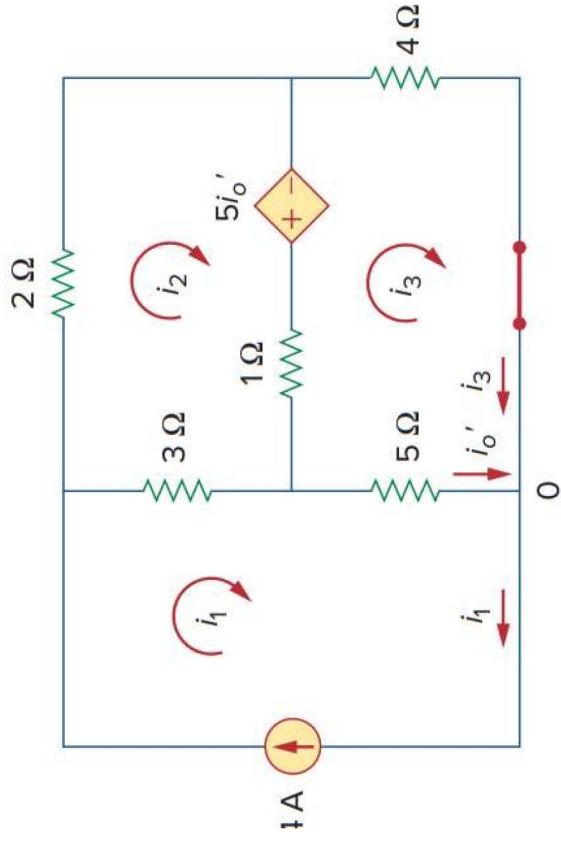
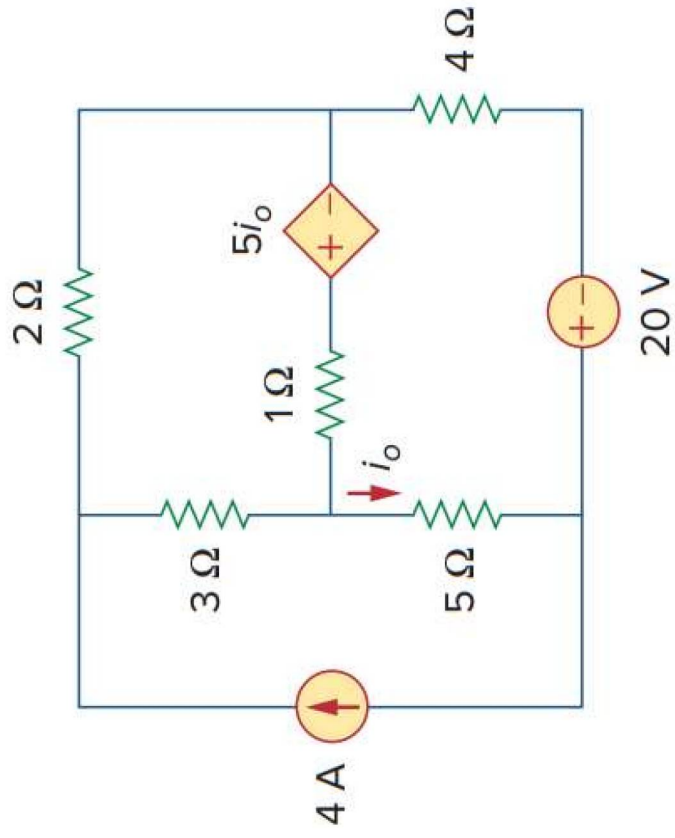
The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone

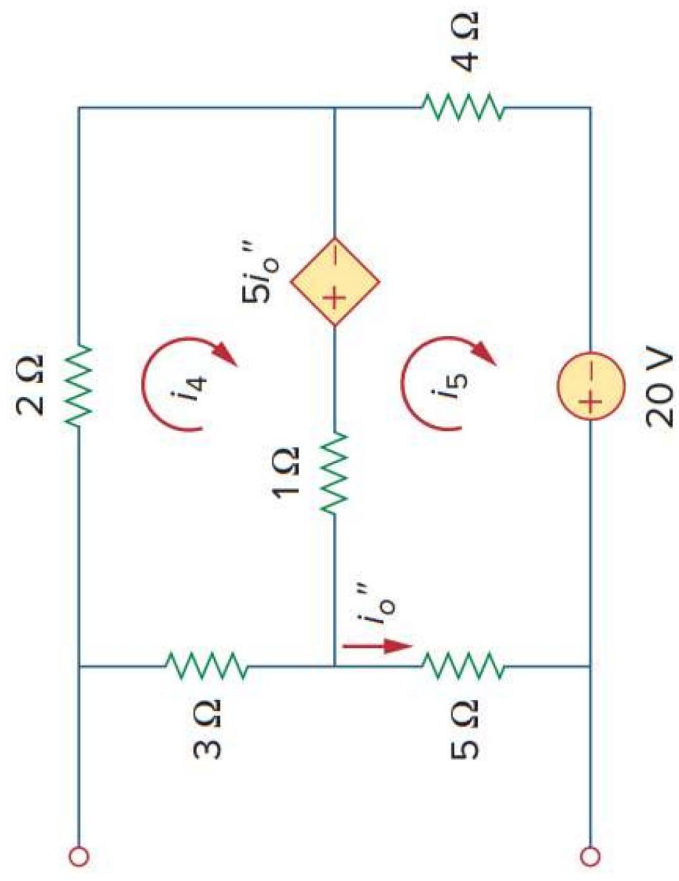
Steps to Apply Superposition Principle:

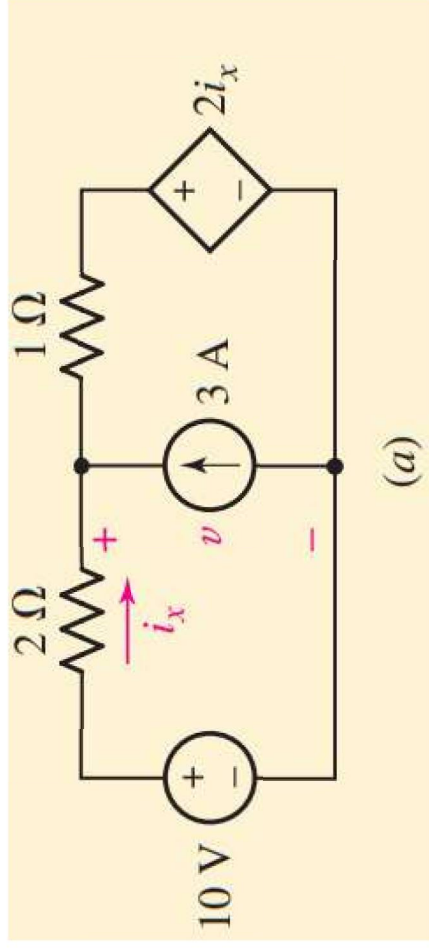
1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using the techniques learned so far (Equivalent resistance, Node analysis, Branch current analysis, Mesh Analysis etc).
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.
4. Dependent sources are left intact because they are controlled by circuit variables.

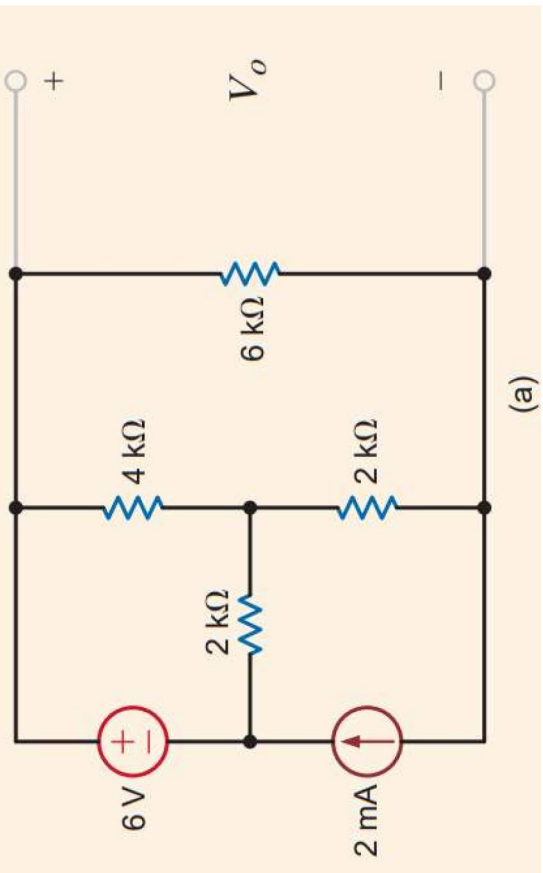
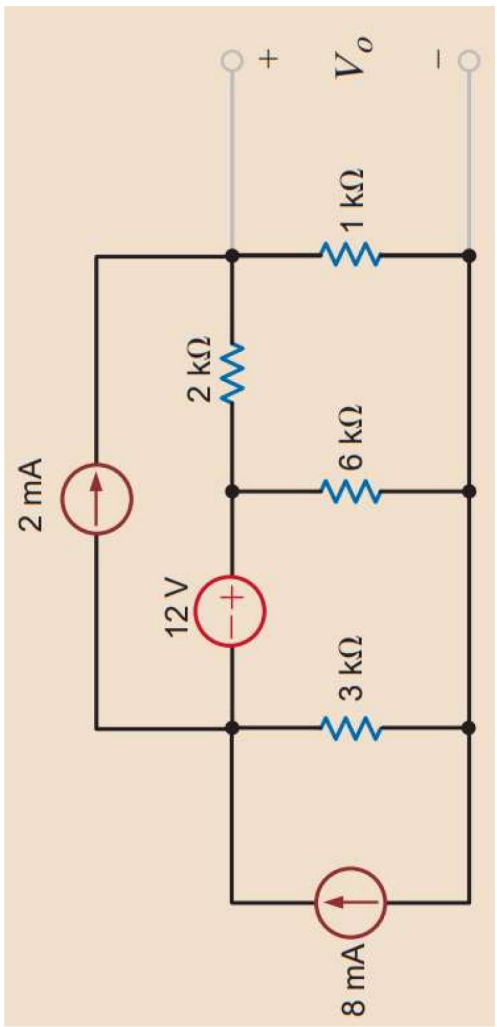


■ **FIGURE 5.2** (a) A voltage source set to zero acts like a short circuit. (b) A current source set to zero acts like an open circuit.









A **source transformation** is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

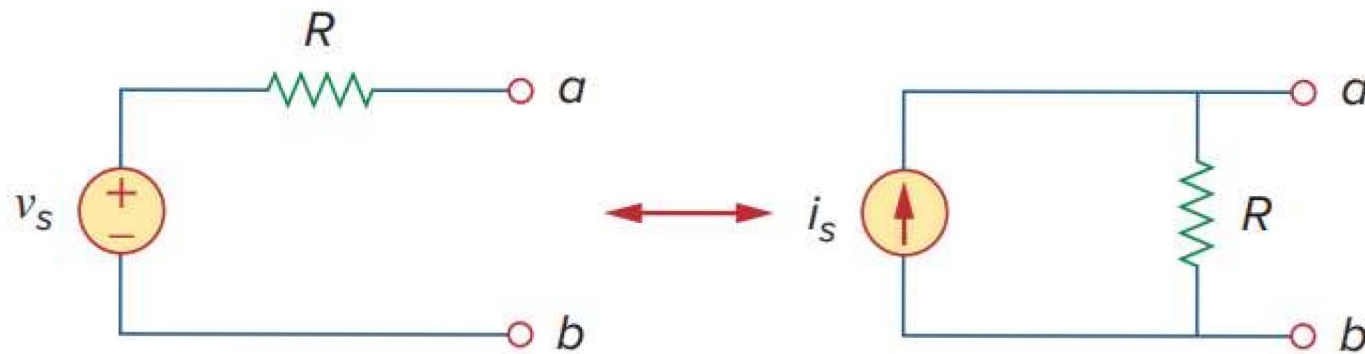


Figure 4.15

Transformation of independent sources.

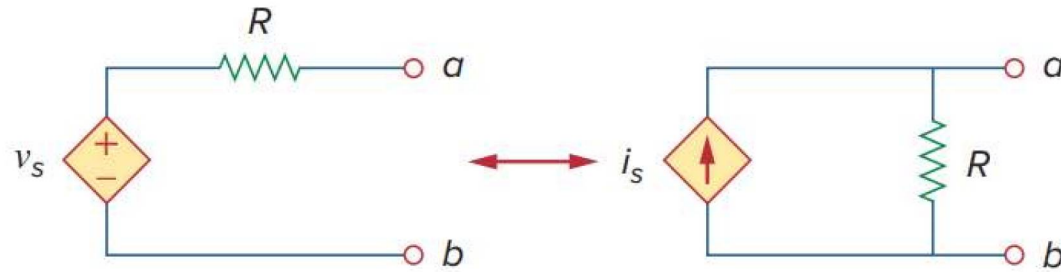
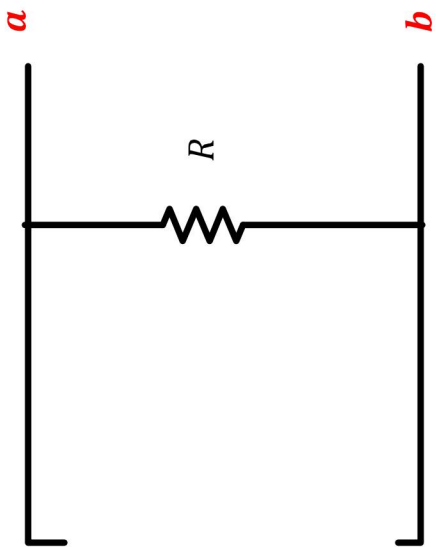
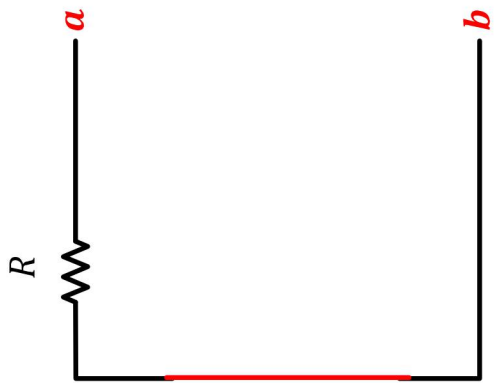
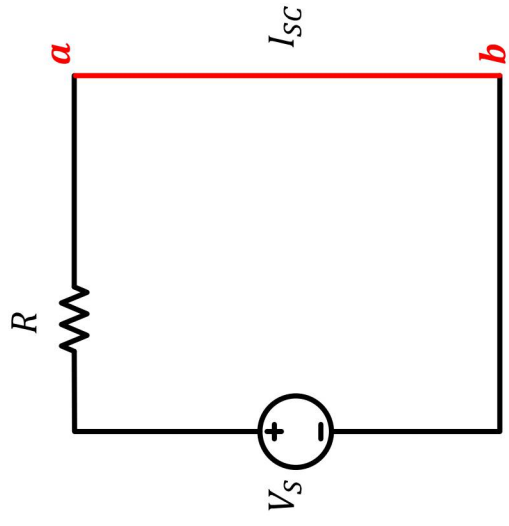


Figure 4.16

Transformation of dependent sources.

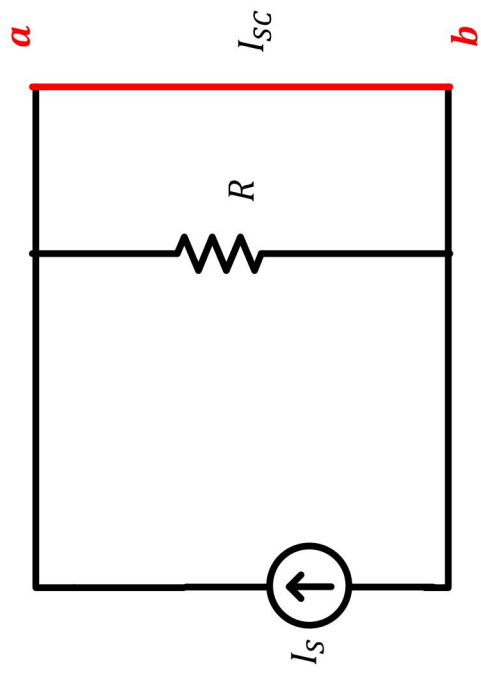
- a source transformation does not affect the remaining part of the circuit.
- the arrow of the current source is directed toward the positive terminal of the voltage source.





$$I_{sc} = \frac{V_s}{R}$$

$$I_s = \frac{V_s}{R}$$



$$I_{sc} = I_s$$

